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Jurusan Sistem Informasi
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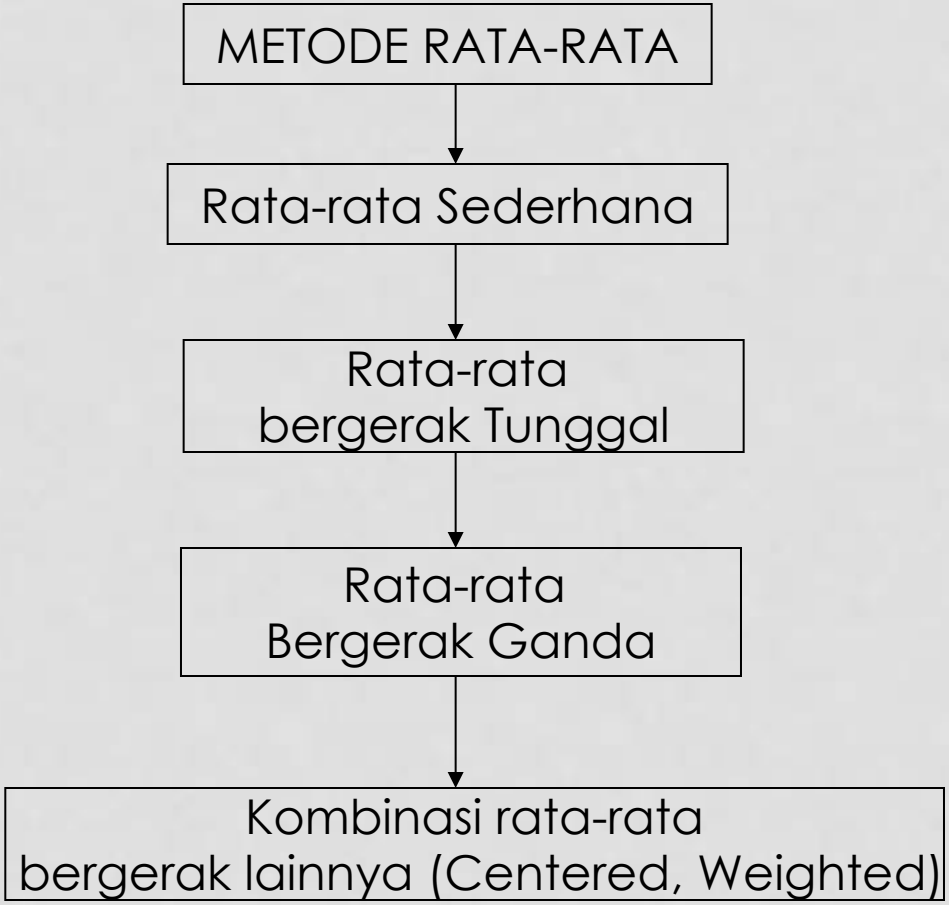
# METODE MOVING AVERAGE

RARAS TYASNURITA

# METODE RATA-RATA (MOVING AVERAGE)

- Definisi :
  - Metode peramalan dengan cara melakukan perhitungan rata-rata dari data sebelumnya untuk meramalkan periode selanjutnya
  - *Averaging methods are suitable for* ***stationary time series data*** *where the series is in equilibrium around a constant value with a constant variance over time.*
  - *This method is appropriate when there is* ***no noticeable trend or seasonality.***

# KLASIFIKASI METODE RATA-RATA



# METODE RATA-RATA SEDERHANA (SIMPLE MOVING AVERAGE)

- The Mean
  - Uses the average of all the historical data as the forecast

$$F_{t+1} = \frac{1}{t}\sum_{i=1}^{t} y_i$$

  - When new data becomes available , the forecast for time t+2 is the new mean including the previously observed data plus this new observation.

$$F_{t+2} = \frac{1}{t+1}\sum_{i=1}^{t+1} y_i$$

$$\text{Moving Average} = \frac{\sum(\text{most recent } n \text{ data values})}{n}$$

# RATA-RATA BERGERAK TUNGGAL ( SINGLE MOVING AVERAGE)

Diberikan N titik data dan diputuskan untuk menggunakan T Observasi pada setiap rata-rata (yang disebut dengan rata-rata bergerak berorde T), atau MA(T), sehingga keadaannya

| Waktu | Rata-rata bergerak | Ramalan |
|---|---|---|
| T | $\bar{x} = \frac{x_1 + x_2 + \ldots + x_T}{T}$ | $F_{T+1} = \bar{x} = \sum_{t=1}^{T} x_i / T$ |
| T+1 | $\bar{x} = \frac{x_2 + x_3 + \ldots + x_{T+1}}{T}$ | $F_{T+2} = \bar{x} = \sum_{t=2}^{T+1} x_i / T$ |
| T+2 | $\bar{x} = \frac{x_3 + x_4 + \ldots + x_{T+2}}{T}$ | $F_{T+3} = \bar{x} = \sum_{t=3}^{T+2} x_i / T$ |

# SINGLE MOVING AVERAGE (2)

- The moving average for time period t is the mean of the "k" most recent observations.
- The constant number k is **specified at the outset.**
- A **large k** is desirable when there are **wide, infrequent fluctuations** in the series.
- A **small k** is most desirable when there are **sudden shifts in the level** of series.
- Each new data point is included in the average as it becomes available, and the oldest data point is discarded.
- For quarterly data, a four-quarter moving average, MA(4), and monthly data, a 12-month moving average, MA(12) eliminates or averages out seasonal effects.

# SINGLE MOVING AVERAGE (3) (STEPS)

1. Determine N
2. Calculate level for all appropriate periods

$$L_t = (D_t + D_{t-1} + \cdots + D_{t-N+1}) / N$$

3. Forecast **(renumber forecast time periods)**

$$F_{t+1} = L_t$$

$$F_{t+n} = L_t$$

What N should be chosen?

Small N makes level very responsive to the last observed demand point

Large N makes level less responsive (smoother)

# SINGLE MOVING AVERAGE (N=3) EXAMPLE 1

| t | Demand | Level | Forecast |
|---|---|---|---|
| 0 | | | |
| 1 | 20 | | |
| 2 | 60 | | |
| 3 | 25 | 35.00 | |
| 4 | 70 | 51.67 | |
| 5 | 28 | 41.00 | |
| 6 | 72 | 56.67 | |
| 1 | | | |
| 2 | | | |

(20+60+25)/3=35

(60+25+70)/3=51.67

(25+70+28)/3=41

(70+28+72)/3=56.67

# SINGLE MOVING AVERAGE (N=3) EXAMPLE 1

| t | Demand | Level | Forecast |
|---|---|---|---|
| 0 | | | |
| 1 | 20 | | |
| 2 | 60 | | |
| 3 | 25 | 35.00 | |
| 4 | 70 | 51.67 | |
| 5 | 28 | 41.00 | |
| 6 | 72 | 56.67 | |
| 1 | | | 56.67 |
| 2 | | | 56.67 |

**Forecast = last level calculated**

# SINGLE MOVING AVERAGE EXAMPLE 2

- Ramalkan penjualan tahun 2007 dengan menggunakan 3 periode pergerakan rata-rata.
  - 2002 4
  - 2003 6
  - 2004 5
  - 2005 3
  - 2006 7
  - 2007 ?

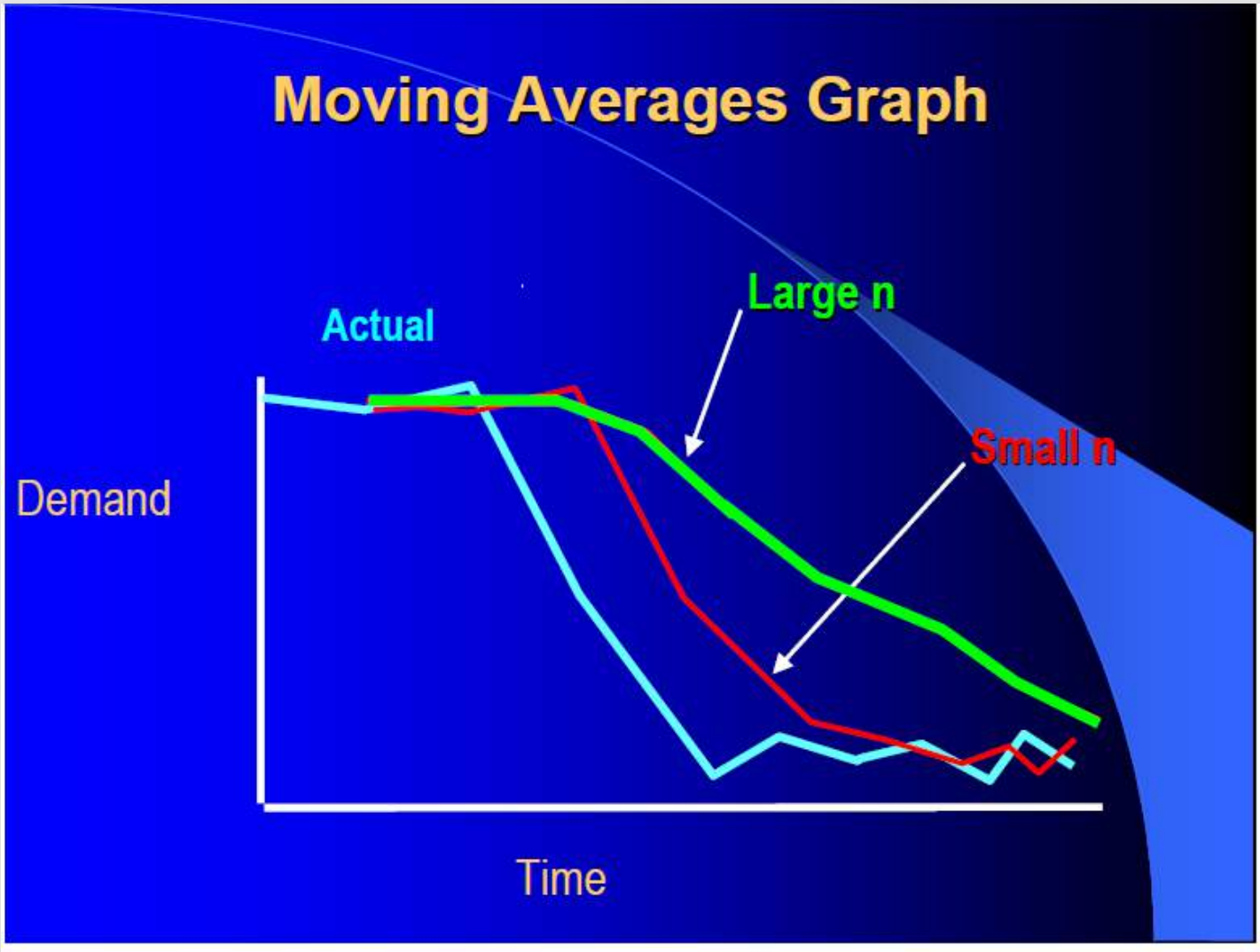
| Time | Response $Y_i$ | Moving Total (n=3) | Moving Average (n=3) |
|---|---|---|---|
| 2002 | 4 | NA | NA |
| 2003 | 6 | NA | NA |
| 2004 | 5 | NA | NA |
| 2005 | 3 | 4+6+5=15 | 15/3=5.0 |
| 2006 | 7 | 6+5+3=14 | 14/3=4.7 |
| 2007 | NA | 5+3+7=15 | 15/3=5.0 |

# SINGLE MOVING AVERAGE EXAMPLE 3

| MONTH | ACTUAL SALES | THREE-MONTH MOVING AVERAGES |
|---|---|---|
| January | 10 | |
| February | 12 | |
| March | 16 | |
| April | 13 | (10 + 12 + 16)/3 = 12.67 |
| May | 17 | (12 + 16 + 13)/3 = 13.67 |
| June | 19 | (16 + 13 + 17)/3 = 15.33 |
| July | 15 | (13 + 17 +19)/3 = 16.33 |
| August | 20 | (17 + 19 + 15)/3 = 17.00 |
| September | 22 | (19 + 15 +20)/3 = 18.00 |
| October | 19 | (15 +20 +22)/3 = 19.00 |
| November | 21 | (20 + 22+ 19)/3 = 20.33 |
| December | 19 | (22 + 19 + 21)/3 = 20.67 |

# SINGLE MOVING AVERAGE EXAMPLE 3

- Time horizon SHORT is better



# SINGLE MOVING AVERAGE
# EXAMPLE 4

Contoh : Rata-rata pengiriman barang

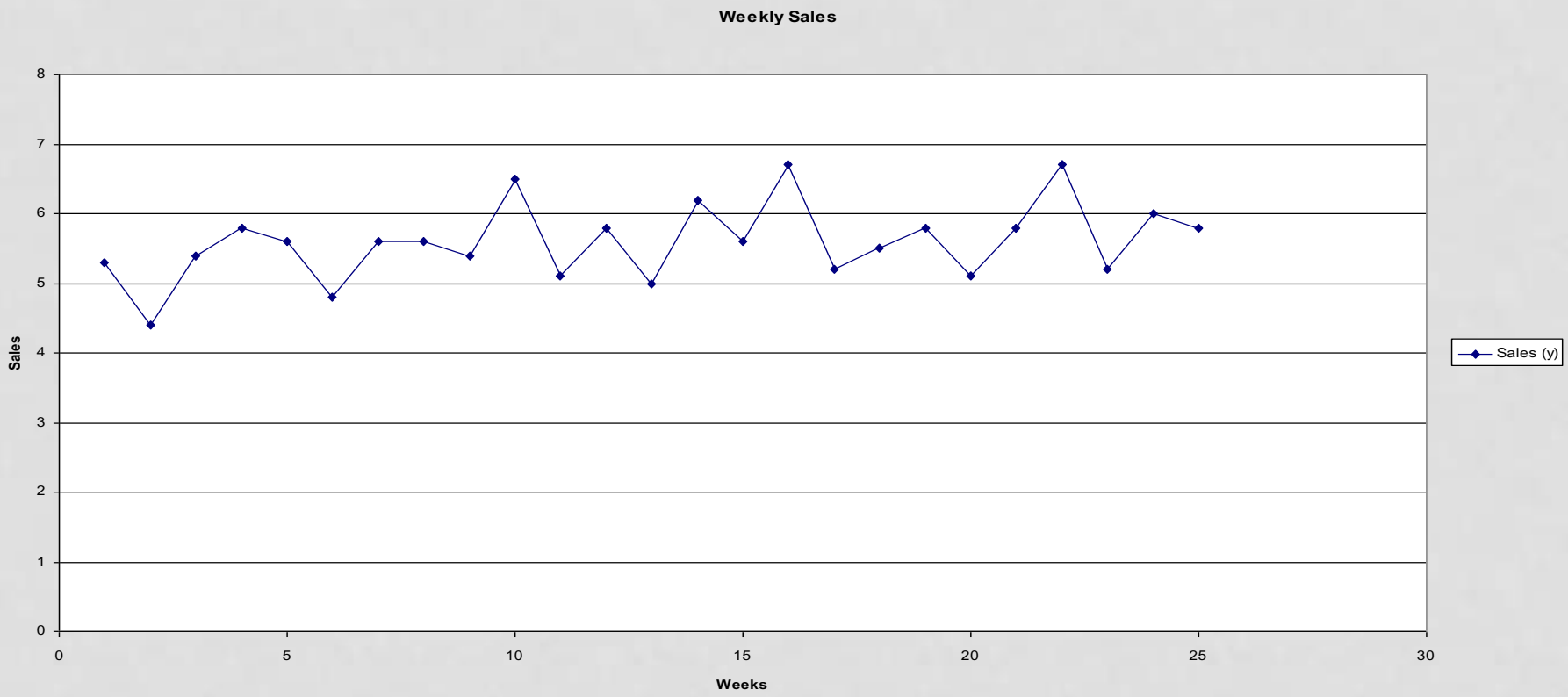
| PERIODE WAKTU | NILAI PENGAMATAN | T=3 | T=5 |
|---|---|---|---|
| 1 | 200 | - | - |
| 2 | 135 | - | - |
| 3 | 195 | - | - |
| 4 | 197.5 | 176.7 | - |
| 5 | 310 | 175.8 | - |
| 6 | 175 | 234.2 | 207.5 |
| 7 | 155 | 227.5 | 202.5 |
| 8 | 130 | 213.3 | 206.5 |
| 9 | 220 | 153.3 | 193.5 |
| 10 | 277 | 168.3 | 198.0 |
| 11 | 235 | 209.2 | 191.4 |
| 12 | - | 244.2 | 203.5 |

## SINGLE MOVING AVERAGE
## EXAMPLE 4 : WEEKLY DEPARTMENT STORE SALES

- The weekly sales figures (in millions of dollars) presented in the following table are used by a major department store to determine the need for temporary sales personnel.

| Period (t) | Sales (y) |
|---|---|
| 1 | 5.3 |
| 2 | 4.4 |
| 3 | 5.4 |
| 4 | 5.8 |
| 5 | 5.6 |
| 6 | 4.8 |
| 7 | 5.6 |
| 8 | 5.6 |
| 9 | 5.4 |
| 10 | 6.5 |
| 11 | 5.1 |
| 12 | 5.8 |
| 13 | 5 |
| 14 | 6.2 |
| 15 | 5.6 |
| 16 | 6.7 |
| 17 | 5.2 |
| 18 | 5.5 |
| 19 | 5.8 |
| 20 | 5.1 |
| 21 | 5.8 |
| 22 | 6.7 |
| 23 | 5.2 |
| 24 | 6 |
| 25 | 5.8 |

# SINGLE MOVING AVERAGE
# EXAMPLE 4 : WEEKLY DEPARTMENT STORE SALES



# SINGLE MOVING AVERAGE
# EXAMPLE 4 : WEEKLY DEPARTMENT STORE SALES

- Use a three-week moving average (k=3) for the department store sales to forecast for the week 24 and 26.
- The forecast for the week 24 is

$$\hat{y}_{24} = \frac{(y_{23} + y_{22} + y_{21})}{3} = \frac{5.2 + 6.7 + 5.8}{3} = 5.9$$

- The forecast error is

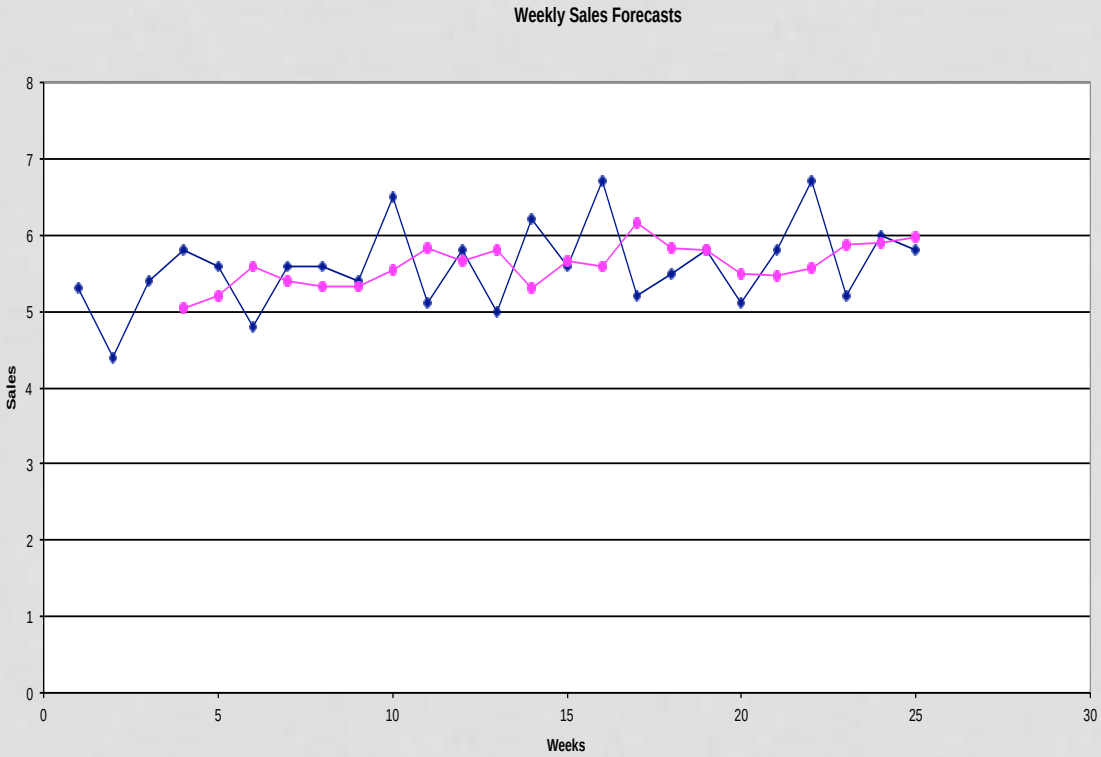
$$e_{24} = y_{24} - \hat{y}_{24} = 6 - 5.9 = .1$$

- The forecast for the week 26 is

$$\hat{y}_{26} = \frac{y_{25} + y_{24} + y_{23}}{3} = \frac{5.8 + 6 + 5.2}{3} = 5.7$$

# SINGLE MOVING AVERAGE
# EXAMPLE 4 : WEEKLY DEPARTMENT STORE SALES

- RMSE = 0.63




| Period (t) | Sales (y) | forecast |
|---|---|---|
| 1 | 5.3 | |
| 2 | 4.4 | |
| 3 | 5.4 | |
| 4 | 5.8 | 5.033333 |
| 5 | 5.6 | 5.2 |
| 6 | 4.8 | 5.6 |
| 7 | 5.6 | 5.4 |
| 8 | 5.6 | 5.333333 |
| 9 | 5.4 | 5.333333 |
| 10 | 6.5 | 5.533333 |
| 11 | 5.1 | 5.833333 |
| 12 | 5.8 | 5.666667 |
| 13 | 5 | 5.8 |
| 14 | 6.2 | 5.3 |
| 15 | 5.6 | 5.666667 |
| 16 | 6.7 | 5.6 |
| 17 | 5.2 | 6.166667 |
| 18 | 5.5 | 5.833333 |
| 19 | 5.8 | 5.8 |
| 20 | 5.1 | 5.5 |
| 21 | 5.8 | 5.466667 |
| 22 | 6.7 | 5.566667 |
| 23 | 5.2 | 5.866667 |
| 24 | 6 | 5.9 |
| 25 | 5.8 | 5.966667 |
| | | 5.666667 |

# RATA-RATA BERGERAK GANDA ( DOUBLE MOVING AVERAGE)

Merupakan rata-rata bergerak dari rata-rata bergerak

| PERIODE | NILAI PENGAMATAN | RAMALAN (N=3) | KESALAHAN |
|---|---|---|---|
| 1 | 2 | - | - |
| 2 | 4 | - | - |
| 3 | 6 | - | - |
| 4 | 8 | 4 | 4 |
| 5 | 10 | 6 | 4 |
| 6 | 12 | 8 | 4 |
| 7 | 14 | 10 | 4 |
| 8 | 16 | 12 | 4 |
| 9 | 18 | 14 | 4 |
| 10 | - | 16 | 4 |

| PERIODE | NILAI AKTUAL | RATA-RATA BERGERAK TUNGGAL (N=3) | KESALAHAN | RATA-RATA BERGERAK GANDA (N=3) | KESALAHAN | RAMALAN +TREN |
|---|---|---|---|---|---|---|
| 1 | 2 | - | - | - | - | - |
| 2 | 4 | - | - | - | - | - |
| 3 | 6 | 4 | 2 | - | - | - |
| 4 | 8 | 6 | 2 | - | - | - |
| 5 | 10 | 8 | 2 | 6 | 2 | - |
| 6 | 12 | 10 | 2 | 8 | 2 | 12 |
| 7 | 14 | 12 | 2 | 10 | 2 | 14 |
| 8 | 16 | 14 | 2 | 12 | 2 | 16 |
| 9 | 18 | 16 | 2 | 14 | 2 | 18 |
| 10 | 20 | 18 | 2 | 16 | 2 | 20 |
| 11 | - | - | - | - | - | 22 |

# LINEAR/DOUBLE MOVING AVERAGE

Prosedur:

1. $SMA_t = S'_t$
2. Perbedaan antara single & double MA(t) = $S'_t - S''_t$
3. Trend periode t ke t+1 (t+m)

Generalisasi:

$$S'_t = \frac{x_t + x_{t-1} + \ldots + x_{t-N+1}}{N}$$

$$S''_t = \frac{s'_t + s'_{t-1} + \ldots + s'_{t-N+1}}{N}$$

$a_t = S'_t + (S'_t - S''_t) = 2S'_t - S''_t$

$b_{tm} = 2/(N-1) * (S'_t - S''_t)$

$b_t = F_{t+m} = a_t + b_{tm}$

# CENTERED MOVING AVERAGE

- Definisi :
  - Centered Moving Average bergerak terpusat terdiri dari komputasi rata-rata data periode n dan menghubungkannya dengan titik tengah periode.
  - Sebagai contoh,

rata-rata untuk periode 5, 6 dan 7 berhubungan dengan periode 6. Hal ini berguna dalam proses komputasi musim indeks.

# CENTERED MOVING AVERAGE (2)

$$\overline{D_t} = \begin{cases} [D_{t-(p/2)} + D_{t+(p/2)} + \text{Sum } 2D_i] \,/\, 2p \textit{ for p even} \\ \quad \text{(sum is from } i = t+1-(p/2) \text{ to } t-1+(p/2)) \\ \\ \text{Sum } D_i \,/\, p \textit{ for p odd} \\ \text{(sum is from } i = t-(p/2) \text{ to } t+(p/2)), \text{ p/2 truncated to lower integer} \end{cases}$$

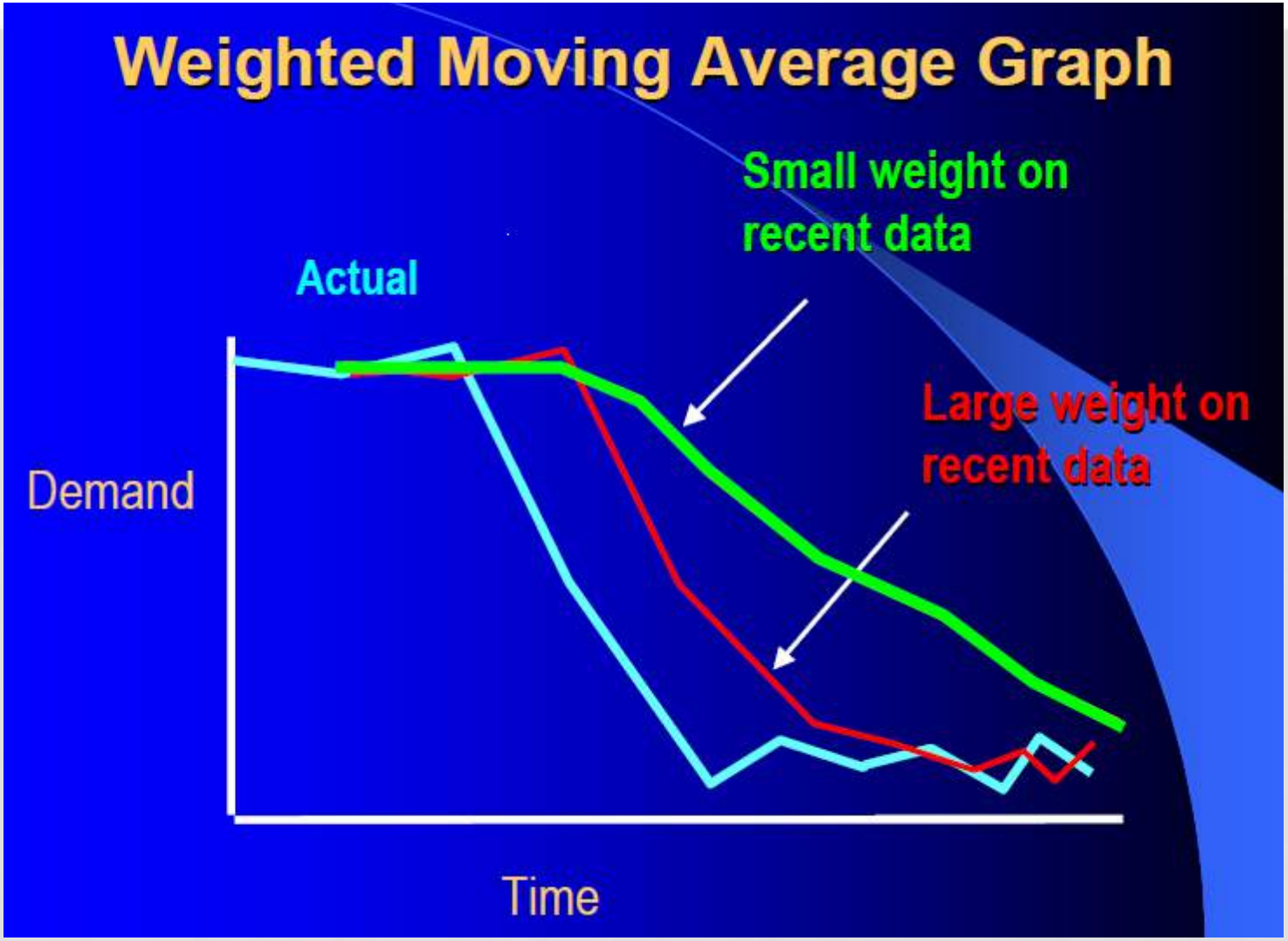
| | | |
|---|---|---|
| 5 | 10 | |
| 6 | 13 | 10+13+11: 3= 11.33 |
| 7 | 11 | |

# WEIGHTED MOVING AVERAGE

- Data yang lebih baru memiliki bobot lebih tinggi dibanding data lama
- Bobot berdasarkan intuisi
  - Biasanya tersusun antara 0 & 1, dan jumlah ke 1.0

$$WMA = \frac{\Sigma(\text{Weight for period n})\,(\text{Value in period n})}{\Sigma\text{Weights}}$$

# WEIGHTED MOVING AVERAGE (2)

| MONTH | ACTUAL SALES | WEIGHTED MOVING AVERAGES |
|---|---|---|
| January | 10 | |
| February | 12 | |
| March | 16 | |
| April | 13 | $(1 \times 10 + 2 \times 12 + 3 \times 16)/6 = 13.67$ |
| May | 17 | $(1 \times 12 + 2 \times 16 + 3 \times 13)/6 = 13.83$ |
| June | 19 | $(1 \times 16 + 2 \times 13 + 3 \times 17)/6 = 15.50$ |
| July | 15 | $(1 \times 13 + 2 \times 17 + 3 \times 19)/6 = 17.33$ |
| August | 20 | $(1 \times 17 + 2 \times 19 + 3 \times 15)/6 = 16.67$ |
| September | 22 | $(1 \times 19 + 2 \times 15 + 3 \times 20)/6 = 18.17$ |
| October | 19 | $(1 \times 15 + 2 \times 20 + 3 \times 22)/6 = 20.17$ |
| November | 21 | $(1 \times 20 + 2 \times 22 + 3 \times 19)/6 = 20.17$ |
| December | 19 | $(1 \times 22 + 2 \times 19 + 3 \times 21)/6 = 20.50$ |

# WEIGHTED MOVING AVERAGE (3)